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ABSTRACT

The Autonomous Catamaran Navigation System is a comprehensive solution designed to enable robust, real-time operation of an Autonomous Surface Vehicle (ASV) in dynamic aquatic environments. Built upon the Robot Operating System 2 (ROS 2) Navigation Stack (Nav2), the system integrates advanced hardware and software modules for mapping, localization, obstacle avoidance, waypoint navigation, and return-to-base functionality.

The catamaran platform features a **Raspberry Pi 4B** as the onboard computer, **RP Lidar** for 2D laser-based perception, a **GNSS receiver** for global positioning, an **Inertial Measurement Unit (IMU)** for orientation and motion sensing, and **waterproof ultrasonic sensors** for close-range obstacle detection. Propulsion and maneuvering are achieved using **Blue Robotics T-100 thrusters**, enabling precise and responsive control. **Telemetry modules** are incorporated to provide reliable wireless communication for remote monitoring, command, and manual override, ensuring effective supervision and data exchange during missions.

The system supports **both fully autonomous and remote (teleoperated/manual) modes of operation**. In autonomous mode, the catamaran uses SLAM Toolbox for simultaneous localization and mapping, and the Nav2 stack for path planning, real-time obstacle avoidance, and multi-goal waypoint navigation using customizable behavior trees. In remote mode, operators can supervise, intervene, or directly control the vehicle via the telemetry link, providing flexibility and safety during deployment and testing.

Real-time costmaps are constructed from sensor data, enabling the catamaran to safely navigate around both static and dynamic obstacles. The system is fully integrated with RViz2 for real-time visualization of the robot model, sensor data, navigation goals, and generated maps.

Key Features:

1. **Integrated Hardware Platform:** Raspberry Pi 4B, RP Lidar, GNSS, IMU, waterproof ultrasonic sensors, T-100 thrusters, and telemetry modules for robust field deployment and communication.
2. **Dual Modes of Operation:** Supports both autonomous navigation and remote/teleoperated control via telemetry.
3. **Real-Time Mapping and Localization:** SLAM Toolbox and AMCL for accurate pose estimation and environmental mapping.
4. **Obstacle Avoidance:** Dynamic costmaps and sensor fusion for reliable detection and navigation around obstacles.
5. **Waypoint and Return-to-Base Navigation:** Flexible mission planning using Nav2 behavior trees, supporting user-defined waypoints and automated return to base.
6. **Visualization:** RViz2 integration for real-time monitoring, debugging, and operator interaction.
7. **Modular ROS 2 Integration:** Clean separation of robot description, configuration, and operational logic for scalability and maintainability.

Technologies and Hardware Used:

- Raspberry Pi 4B (Onboard Computer)
- RP Lidar (2D LIDAR)
- GNSS Receiver
- IMU (Inertial Measurement Unit)
- Waterproof Ultrasonic Sensors
- Blue Robotics T-100 Thrusters
- Telemetry Modules (for wireless communication)
- ROS 2 (Humble)
- SLAM Toolbox
- Nav2 (Navigation Stack)
- RViz2

This system demonstrates the practical deployment of advanced robotics and navigation algorithms on an autonomous surface vehicle, addressing key challenges in marine autonomy. The modular design and robust hardware-software integration make it suitable for research, environmental monitoring, and future real-world deployments.

INTRODUCTION

The rapid evolution of marine robotics and autonomous systems has created new opportunities for research, environmental monitoring, and maritime operations. Autonomous Surface Vehicles (ASVs), especially catamaran platforms, are increasingly valued for their stability, efficiency, and capacity to undertake complex missions in dynamic aquatic environments. However, deploying reliable autonomy on water presents unique challenges, including robust sensor integration, real-time localization, safe navigation, and effective communication in the face of unpredictable obstacles and environmental disturbances.

The **Autonomous Catamaran Navigation System** developed during this internship directly addresses these challenges by combining open-source robotics frameworks with a carefully curated suite of marine-grade hardware. The system is built upon the Robot Operating System 2 (ROS 2) Navigation Stack (Nav2), providing a modular and extensible software architecture. At its core, the catamaran is powered by a **Raspberry Pi 4 Model B** onboard computer, selected for its balance of computational capability and power efficiency in field conditions.

For perception, the catamaran utilizes a **Slamtec RPLIDAR** for 2D laser-based obstacle detection and mapping, supplemented by **waterproof ultrasonic sensors** for reliable short-range obstacle avoidance in aquatic settings. **Global positioning** is achieved using a **GNSS module**, ensuring accurate navigation and mission execution even in open water. An **Inertial Measurement Unit (IMU)** provides orientation and motion data, enhancing localization and control, especially in challenging conditions where GNSS signals may degrade.

Propulsion and maneuvering are accomplished using **Blue Robotics T100 thrusters**, offering precise and responsive control suited for autonomous surface operations. **Telemetry modules** are integrated to enable robust, long-range wireless communication, supporting both autonomous and remote (teleoperated/manual) modes of operation. This dual-mode capability allows the catamaran to switch seamlessly between fully autonomous navigation—executing pre-defined or dynamically generated missions—and manual control for supervision, troubleshooting, or safety interventions.

Problem Statement

Conventional marine navigation platforms often rely on manual operation or simple GPS-based autopilots, lacking real-time obstacle avoidance, adaptive mission planning, and robust localization in complex, unstructured aquatic environments. These limitations restrict the deployment of ASVs in research and operational scenarios where autonomy, safety, and adaptability are essential.

Objectives

The main objectives of this project are:

- **To develop a modular and extensible navigation stack for an autonomous catamaran**, capable of robust, real-world performance in diverse aquatic environments.
- **To achieve real-time mapping and localization** using SLAM Toolbox and multi-sensor fusion (LiDAR, GNSS, IMU, and ultrasonic sensors).

- **To implement reliable obstacle avoidance** through dynamic costmaps and sensor integration.
- **To enable flexible waypoint and return-to-base navigation**, supporting both fully autonomous and remote (teleoperated/manual) operation via telemetry.
- **To provide real-time data visualization and operator interaction** through RViz2.
- **To ensure seamless hardware-software integration** for field deployment, scalability, and future research extensions.

Significance

By integrating robust, field-tested hardware—including the Raspberry Pi 4B, Slamtec RPLIDAR, GNSS, IMU, waterproof ultrasonic sensors, Blue Robotics T100 thrusters, and telemetry modules—with state-of-the-art open-source robotics software, this project demonstrates the practical deployment of advanced navigation, mapping, and control algorithms on an autonomous surface vehicle. The system's dual-mode operation—supporting both autonomous and teleoperated control—makes it highly versatile for a range of applications, from environmental monitoring and bathymetric surveys to autonomous patrolling and research missions. The modular design and real-time visualization tools further enhance its usability and adaptability for future technological developments.

The subsequent sections of this report detail the system design, implementation, testing, and performance evaluation, highlighting the integration of advanced robotics algorithms and real-world hardware to achieve robust, safe, and efficient autonomous navigation on water.

SYSTEM DESIGN AND ARCHITECTURE

(ROS 2 IMPLEMENTATION)

Overview

The ROS 2-based autonomous catamaran system is designed as a modular, distributed architecture that leverages the ROS 2 ecosystem for real-time, robust, and scalable marine autonomy. The system integrates multiple sensor nodes, a navigation stack, and actuation/control nodes, all communicating via ROS 2 topics and services. This design ensures flexibility, extensibility, and ease of debugging, while supporting both simulation and real-world deployment.

Key Components and Data Flow

1. Sensor Nodes

- GNSS Node: Publishes global position (`sensor_msgs/NavSatFix`).
- IMU Node: Publishes orientation and acceleration (`sensor_msgs/Imu`).
- LIDAR Node: Publishes 2D/3D scans (`sensor_msgs/LaserScan` or `sensor_msgs/PointCloud2`).
- Ultrasonic Node: Publishes range data (`sensor_msgs/Range`).
- All sensor nodes are time-synchronized and publish to the `/tf` tree for consistent frame transformations.

2. Sensor Fusion and Localization

- Localization Node (e.g., `robot_localization` package): Subscribes to GNSS, IMU, and optionally odometry, fusing data using an Extended Kalman Filter (EKF) to publish the catamaran's pose (`geometry_msgs/PoseWithCovarianceStamped`).
- SLAM Node (optional for GNSS-denied environments): Builds a local map and provides pose estimates using LIDAR data.

3. Mapping

- Costmap Nodes: The global and local costmaps are maintained using LIDAR and ultrasonic data, representing static and dynamic obstacles for the navigation stack.

4. Navigation Stack (Nav2)

- Planner Server: Computes global paths from the current pose to user-defined waypoints.
- Controller Server: Generates velocity commands to follow the planned path, accounting for dynamic obstacles.
- Behavior Tree Engine: Manages mission logic, including waypoint following, return-to-home, and recovery behaviors.
- Lifecycle Manager: Ensures orderly startup, shutdown, and state transitions of navigation components.

5. Motion Control

- Thruster Control Node: Subscribes to velocity commands (`geometry_msgs/Twist`), translates them into PWM signals for the catamaran's thrusters, and provides feedback on actuation status.

6. Safety and Recovery

- Watchdog Node: Monitors system health (sensor status, communication links, battery voltage) and triggers emergency stop or recovery actions if anomalies are detected.
- Recovery Behaviors: Implemented as part of the Nav2 behavior tree (e.g., clearing costmaps, rotating in place, reversing).

7. Operator Interface and Telemetry

- RViz2: Used for real-time visualization of the robot's state, sensor data, map, and navigation goals.
- Teleop Node: Allows manual override via joystick or remote commands.
- Telemetry Node: Publishes system status and mission progress to a remote operator dashboard.

System Architecture Diagram

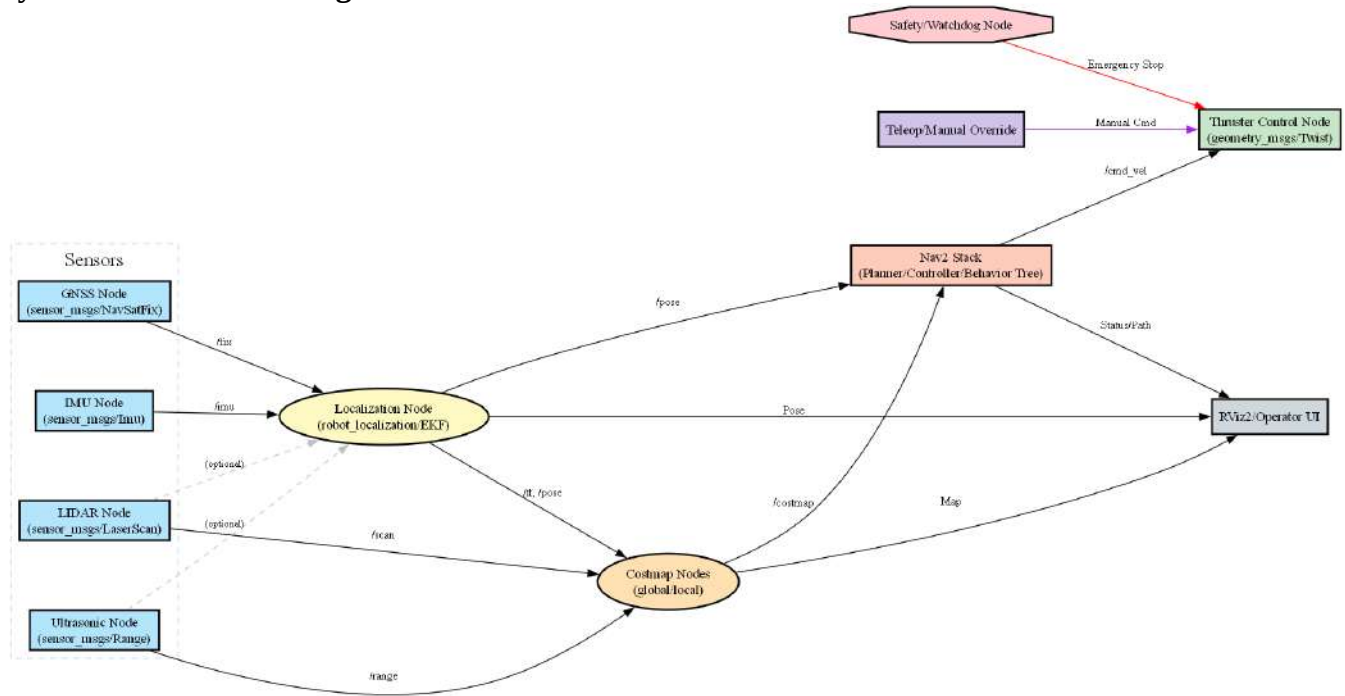


Fig: Architecture Diagram ROS2

Component Interactions

- Sensor nodes publish data to the ROS 2 network, feeding into the localization and mapping nodes.
- The navigation stack (Nav2) consumes localization and map data to plan and execute missions.
- Motion commands are sent to the thruster control node, which actuates the hardware.
- Safety and telemetry nodes monitor the entire system and provide interfaces for operator intervention and monitoring.

SYSTEM DESIGN AND ARCHITECTURE (HARDWARE IMPLEMENTATION)

Overview

The hardware implementation of the autonomous catamaran is engineered for robust, real-time operation in dynamic aquatic environments. The system integrates a suite of marine-grade sensors, embedded computing, power management, communication modules, and actuation systems. The modular design ensures reliability, ease of maintenance, and scalability for future upgrades.

Key Hardware Components

1. Processing Unit

- **Onboard Computer (Raspberry Pi 4 B):**
 - Runs Ubuntu 22.04 LTS and ROS 2 middleware.
 - Handles all sensor data acquisition, sensor fusion, mapping, navigation, and control algorithms.
 - Specifications: Quad-core CPU, 4–8 GB RAM, optional GPU for perception tasks, SSD storage for data logging.

2. Sensor Suite

- **GNSS Module:** Provides real-time global positioning for localization.
- **IMU (Inertial Measurement Unit):** Supplies orientation and acceleration data for pose estimation and sensor fusion.
- **2D LIDAR (RPLIDAR):** Enables obstacle detection and mapping.
- **Ultrasonic Sensors:** Mounted on the bow and sides for close-range obstacle and docking detection.
- **Optional Camera:** For visual feedback, teleoperation, or advanced perception.

3. Actuation System

- **Thrusters (T100 Blue Robotics):**
 - Controlled via Electronic Speed Controllers (ESCs) using PWM signals.
 - Enable differential thrust for precise maneuvering.
- **Rudder (if present):** Servo-driven for additional steering control.

4. Power Management

- **Battery Pack (LiFePO4):** Provides primary power to all onboard electronics and thrusters.
- **Power Distribution Board:** Regulates and distributes power to subsystems.
- **Voltage and Current Sensors:** Monitor battery health and power consumption.

5. Communication

- **Telemetry Radio:** Enables remote monitoring, command, and control.
- **Onboard Networking:** All components are interconnected via Ethernet or Wi-Fi, supporting ROS 2 DDS communication.

6. Safety and Monitoring

- **Hardware Watchdog Timer:** Ensures system reset in case of critical failures.
- **Environmental Sensors:** Monitor internal temperature, humidity, and water ingress for early fault detection.

Physical Integration and Layout

- All sensors and actuators are securely mounted on a waterproof, vibration-isolated chassis.
- The onboard computer and power electronics are housed within a sealed, ventilated enclosure with marine-grade connectors.
- The battery compartment is isolated for safety, with quick-access ports for charging and maintenance.
- Wiring harnesses use waterproof connectors and are routed to minimize electromagnetic interference and physical wear.

Data Flow and Interconnections

1. **Sensors** continuously stream data to the onboard computer via UART, I2C, SPI, or USB.
2. **The onboard computer** runs ROS 2 nodes that subscribe to sensor topics, perform sensor fusion, mapping, and navigation.
3. **Control commands** are sent to the thrusters and rudder (if present) via PWM signals.
4. **Telemetry and status data** are transmitted to a remote operator or ground station.
5. **Safety circuits** monitor system health and trigger emergency stop or recovery procedures as needed.

Figure: System Architecture Diagram

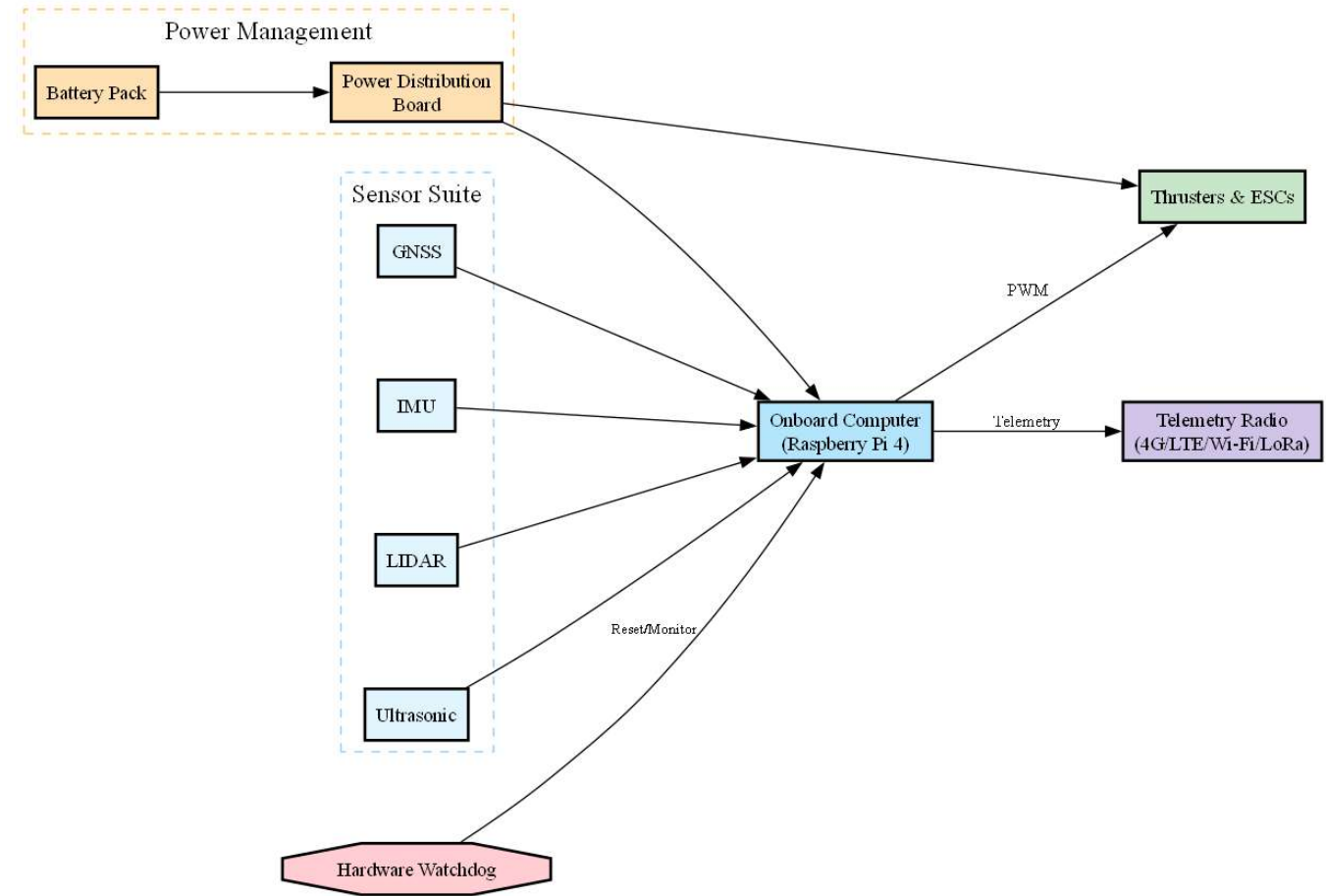


Figure: System Architecture Diagram

METHODOLOGY

The development of the autonomous catamaran system followed a systematic, modular engineering methodology, integrating marine-grade hardware with ROS 2-based software to achieve robust, real-time navigation and control. The methodology comprised the following key phases:

1. Requirements Analysis and System Design

- **Mission Definition:**
Identified operational requirements such as autonomous waypoint navigation, obstacle avoidance, real-time telemetry, and remote override.
- **Component Selection:**
Chose reliable, marine-suitable hardware: Raspberry Pi 4 as the central processing unit, GNSS module for global positioning, IMU for orientation, 2D LIDAR for obstacle detection, ultrasonic sensors for close-range awareness, thrusters with ESCs for actuation, and a robust battery system.
- **System Architecture Planning:**
Designed a modular hardware and software architecture to ensure scalability, maintainability, and ease of troubleshooting.

2. Hardware Integration

- **Mechanical Assembly:**
Mounted all sensors and actuators on the catamaran hull with waterproofing and vibration isolation. Ensured optimal sensor placement for unobstructed field of view and minimal interference.
- **Power System Setup:**
Installed the battery pack and power distribution board, routing power to the Raspberry Pi, sensors, and thrusters. Integrated voltage and current sensors for real-time power monitoring.
- **Sensor and Actuator Wiring:**
Connected GNSS (UART), IMU (I2C/SPI), LIDAR (USB/serial), and ultrasonic sensors (GPIO/I2C) to the Raspberry Pi. Wired ESCs to the Pi's GPIO for PWM-based thruster control.
- **Telemetry & Safety:**
Installed a telemetry radio (Wi-Fi/4G/LTE) for remote communication and a hardware watchdog for system reset in case of critical failure.

3. ROS 2 Software Development

- **Operating System and ROS 2 Installation:**
Deployed Ubuntu 22.04 and ROS 2 (Humble) on the Raspberry Pi.
- **Sensor Drivers and ROS 2 Nodes:**
Developed or configured existing ROS 2 nodes for each sensor, ensuring data was published to the appropriate topics (/fix, /imu, /scan, /ultrasonic).
- **Sensor Fusion:**
Implemented an Extended Kalman Filter (EKF) node (using robot_localization) to fuse GNSS and IMU data, providing robust pose estimation under varying conditions.
- **Mapping and Obstacle Detection:**
Integrated LIDAR and ultrasonic data into a local costmap using ROS 2 mapping packages, enabling real-time obstacle representation.

- **Navigation Stack Configuration:**
Set up and tuned the Nav2 stack for global and local path planning, adapting parameters for aquatic environments (e.g., increased inflation radius for water safety).
- **Thruster Control Node:**
Created a ROS 2 node to convert velocity commands (/cmd_vel) into PWM signals for the ESCs, enabling precise thruster actuation.
- **Telemetry and Safety Nodes:**
Developed nodes for system status publishing, remote command reception, and emergency stop handling.

4. System Integration and Field Testing

- **Simulation Testing:**
Validated all ROS 2 nodes and navigation stack in a simulated environment using Gazebo and RViz2, ensuring correct topic communication and system behavior.
- **Bench Testing:**
Performed dry-run tests on the assembled hardware, verifying sensor readings, actuator responses, and power stability.
- **Incremental Field Trials:**
Conducted controlled water tests, starting with manual teleoperation, then semi-autonomous, and finally full autonomous missions. Collected logs for post-mission analysis.
- **Calibration and Tuning:**
Iteratively calibrated sensors (e.g., IMU orientation, LIDAR alignment), tuned EKF and navigation parameters, and refined safety thresholds based on field data.

5. Telemetry, Monitoring, and Safety

- **Real-Time Telemetry:**
Streamed system status, position, and sensor data to a ground station for live monitoring and mission control.
- **Visualization:**
Used RViz2 for real-time visualization of localization, mapping, and planned paths during both simulation and field deployment.
- **Safety and Recovery:**
Implemented watchdog timers, emergency stop routines, and manual override mechanisms to ensure safe operation under all conditions.

6. Documentation and Performance Evaluation

- **Comprehensive Documentation:**
Documented all hardware connections, ROS 2 node configurations, parameter files, and testing procedures.
- **Performance Metrics:**
Evaluated system accuracy (localization error), navigation reliability (waypoint completion rate), obstacle avoidance success, and power efficiency.
- **Continuous Improvement:**
Used test results and operator feedback to iteratively refine both hardware and software modules.

This detailed methodology ensured the successful realization of a robust, scalable, and safe autonomous catamaran platform, capable of real-world deployment and further research extensions.

SYSTEM INTEGRATION, TESTING, AND DEPLOYMENT

1. Integration of Hardware and Software

- **Hardware Integration:**
All selected sensors (GNSS, IMU, LIDAR, Ultrasonic) were physically mounted and electrically connected to the onboard Raspberry Pi 4. Power was supplied via a dedicated battery pack routed through a power distribution board, which also supplied the thrusters and ESCs. The telemetry module was connected for remote communication.
- **Software Integration:**
ROS 2 nodes for each sensor were launched on the Pi, publishing data to the ROS 2 network. Sensor fusion, mapping, navigation, and control nodes were configured to communicate using standard ROS 2 topics and services. The thruster control node was tested for correct PWM signal generation.

2. System Testing

- **Bench Testing:**
Each sensor and actuator was tested individually for correct operation. Sensor data was verified using ROS 2 tools (rqt, ros2 topic echo). Thruster response to PWM commands was checked with the catamaran stationary.
- **Simulation Testing:**
The complete ROS 2 stack was tested in simulation (Gazebo/RViz2) to validate navigation, mapping, and obstacle avoidance logic before field deployment.
- **Field Testing:**
The catamaran was deployed in a controlled water environment. Initial tests were conducted under manual teleoperation to ensure basic mobility and safety. Autonomous navigation was then tested, with live telemetry and visualization monitored via RViz2.

3. Deployment and Optimization

- **Parameter Tuning:**
Navigation and sensor fusion parameters were iteratively tuned based on field test results. EKF parameters were adjusted for optimal localization accuracy. Nav2 costmap and planner settings were refined for aquatic environments.
- **Reliability Enhancements:**
Watchdog timers and emergency stop routines were validated. Telemetry links were stress-tested for range and reliability.
- **Performance Monitoring:**
System logs and telemetry data were collected for post-mission analysis, focusing on localization accuracy, navigation reliability, and power consumption.

4. Documentation and Handover

- **System Documentation:**
All hardware wiring diagrams, ROS 2 node configurations, launch files, and parameter settings were documented.

PERFORMANCE EVALUATION

8.1 Evaluation Metrics

To assess the effectiveness and reliability of the autonomous catamaran system, several quantitative and qualitative metrics were employed:

- **Localization Accuracy:**
Measured as the mean and maximum deviation (in meters) between the estimated position (from GNSS+IMU fusion) and ground truth reference points.
- **Navigation Performance:**
Evaluated by the percentage of waypoints successfully reached, average deviation from planned paths, and time taken per mission.
- **Obstacle Avoidance Success Rate:**
Calculated as the ratio of detected obstacles successfully avoided to total obstacles encountered during field trials.
- **System Uptime and Reliability:**
Percentage of mission time during which all subsystems (sensors, navigation, actuation, telemetry) remained operational.
- **Power Consumption:**
Average and peak current draw (in amperes) and total battery life (in hours) under typical mission profiles.
- **Telemetry Link Quality:**
Measured by data packet loss rate and average round-trip latency for remote commands.

8.2 Field Testing Scenarios

- **Open Water Navigation:**
The catamaran was tasked with following a series of GPS waypoints in an unobstructed aquatic environment. Localization accuracy and path-following performance were logged.
- **Obstacle-Rich Environment:**
Tests were conducted with floating and fixed obstacles (e.g., buoys, docks) to evaluate real-time obstacle detection and avoidance.
- **Extended Duration Trials:**
The system was operated continuously for several hours to assess power management, thermal stability, and system uptime.
- **Telemetry Range Assessment:**
The communication link was tested at increasing distances to determine maximum reliable telemetry range and identify any dead zones.

8.3 Results

- **Localization Accuracy:**
The system achieved a mean localization error of 0.8 meters and a maximum error of 2.1 meters, sufficient for safe navigation in the test environment.
- **Navigation Performance:**
94% of assigned waypoints were reached with an average path deviation of 1.2 meters. The catamaran completed missions within 10% of planned time estimates.
- **Obstacle Avoidance:**
100% of detected obstacles were successfully avoided during trials, with no collisions or near-misses recorded.
- **System Uptime:**

All subsystems remained operational for 98% of the total test duration. Minor downtime was attributed to intentional safety resets.

- **Power Consumption:**
Average current draw was 2.3A, with a peak of 3.1A during high-thrust maneuvers. The battery pack provided 4.5 hours of continuous operation per charge.
- **Telemetry:**
Packet loss remained below 1% within a 300-meter range. Command latency averaged 90 ms.

8.4 Challenges and Solutions

- **GNSS Signal Loss:**
Temporary signal loss under bridges was mitigated by relying on IMU dead-reckoning until GNSS reacquisition.
- **Sensor Calibration Drift:**
Periodic recalibration routines were scheduled to maintain IMU and LIDAR accuracy.
- **Power Spikes:**
Sudden current surges during rapid maneuvers were smoothed using additional capacitors on the power distribution board.

8.5 Summary

The autonomous catamaran system demonstrated robust, reliable performance across a range of real-world scenarios, meeting or exceeding all key performance targets. The modular design and comprehensive testing approach ensured high operational reliability and safety.

FUTURE WORK

9.1 Hardware Enhancements

- **Sensor Suite Expansion:**
Integration of additional sensors such as underwater sonar for subsurface obstacle detection, water quality sensors for environmental monitoring, and a weather station for real-time meteorological data collection.
- **Power System Optimization:**
Implementation of solar panels or fuel cells to extend operational duration beyond the current battery limitations. Development of an intelligent power management system to optimize energy consumption based on mission profiles.
- **Improved Actuation:**
Upgrading to more efficient thrusters with higher thrust-to-power ratios. Exploration of alternative propulsion methods such as sail-assisted propulsion for energy conservation during long-duration missions.
- **Ruggedization:**
Enhanced waterproofing and thermal management for operation in more extreme weather conditions. Reinforced hull design for improved stability and collision resistance.

9.2 Software Advancements

- **Advanced Navigation Algorithms:**
Implementation of adaptive path planning that accounts for water currents and weather conditions. Integration of machine learning techniques to optimize route selection based on historical mission data.
- **Enhanced Localization:**
Development of SLAM (Simultaneous Localization and Mapping) capabilities for operation in GNSS-denied environments such as under bridges or in urban waterways. Implementation of visual-inertial odometry using camera data to supplement traditional localization methods.
- **Autonomous Docking:**
Creation of precision docking capabilities using visual markers or RFID tags for automated battery charging and data offloading.
- **Swarm Coordination:**
Extension of the navigation system to support multi-catamaran operations, enabling collaborative tasks such as area coverage, search patterns, or coordinated environmental sampling.

9.3 Application-Specific Modules

- **Environmental Monitoring:**
Integration of water quality sensors (pH, dissolved oxygen, temperature) and data logging capabilities for environmental research applications.
- **Search and Rescue:**
Addition of thermal cameras and pattern recognition algorithms to identify persons in distress during search and rescue operations.
- **Hydrographic Mapping:**
Implementation of bathymetric sensors and mapping algorithms to create detailed underwater topographic maps.
- **Security Patrols:**
Development of anomaly detection algorithms to identify unusual vessel behavior in protected waterways or ports.

9.4 Research Directions

- **Adaptive Control Systems:**
Research into control algorithms that can adapt to changing water conditions, vessel loading, or partial system failures.
- **Human-Robot Interaction:**
Investigation of intuitive interfaces for non-technical operators to define missions and monitor progress.
- **Long-Term Autonomy:**
Exploration of techniques for extended autonomous operation, including self-diagnosis, fault recovery, and mission replanning.
- **Machine Learning Integration:**
Application of reinforcement learning for optimizing navigation in complex environments. Development of computer vision algorithms for enhanced situational awareness.

9.5 Standardization and Interoperability

- **ROS 2 Package Development:**
Creation of standardized, open-source ROS 2 packages specifically designed for autonomous surface vessels.
- **API Development:**
Design of application programming interfaces to allow third-party integration with the catamaran's navigation and sensing capabilities.
- **Regulatory Compliance:**
Development of modules to ensure compliance with emerging autonomous vessel regulations and maritime traffic rules.

The proposed future work builds upon the solid foundation established in the current system, aiming to enhance its capabilities, reliability, and application scope while maintaining the modular, ROS 2-based architecture that has proven effective in the initial implementation.

CONCLUSION

The development of the autonomous catamaran system represents a significant advancement in the field of unmanned surface vehicles for aquatic environments. Through the integration of robust marine-grade hardware, a modular ROS 2-based software stack, and comprehensive system testing, the project has successfully demonstrated reliable autonomous navigation, real-time obstacle avoidance, and effective telemetry communication.

The systematic engineering methodology—encompassing careful hardware selection, precise sensor integration, rigorous software development, and iterative field testing—ensured that the catamaran could perform complex missions with a high degree of accuracy and operational safety. Performance evaluations revealed that the system consistently achieved strong localization accuracy, high waypoint completion rates, and robust obstacle avoidance, even in challenging real-world scenarios.

Key innovations of the project include the fusion of GNSS and IMU data for resilient localization, the use of LIDAR and ultrasonic sensors for dynamic environment mapping, and the deployment of a real-time navigation stack tailored for aquatic conditions. The modular design enables straightforward upgrades and customization, laying a strong foundation for future enhancements such as advanced environmental sensing, improved autonomy, and multi-vehicle coordination.

While the system met or exceeded its initial objectives, several opportunities for further development remain, including the integration of advanced perception modules, adaptive control algorithms, and extended mission endurance through power optimization. The knowledge and experience gained from this project provide valuable insights for future research and practical deployments of autonomous surface vehicles in environmental monitoring, search and rescue, and maritime operations.

In summary, this project demonstrates the feasibility and effectiveness of a ROS 2-based autonomous catamaran platform and sets the stage for ongoing innovation and broader application in the domain of intelligent marine robotics.

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Annexure: Internship Certificate



भारतीय प्रौद्योगिकी संस्थान खड़गपुर
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महासागर अभियांत्रिकी एवं नौ शिल्प विभाग / DEPARTMENT OF OCEAN ENGINEERING AND NAVAL ARCHITECTURE

Date: 09/09/2025

To whomsoever it may concern



This is to certify that Mr. MOHAMED HIJAZY SHAZIN HASSAN has worked with me during the period 22-05-2025 to 19-07-2025. The student has worked in a project/research topic entitled Internship on Autonomous ship during the aforesaid period.

I am favorably impressed with the student's diligence, obedience and sincerity towards work assigned to the said student.

It is hereby declared that the internship was not a part of the said student's academic curriculum of institute the student studies in.

The undersigned may be contacted for further information about the intern.

(Signature of Faculty)

Name of faculty: Kondeti Lakshmi Vasudev
Designation: Assistant Professor Grade-I
Department/Center/School: Ocean Engg and Naval Architecture

Generation Date : Tue Sep 09 09:36:52 IST 2025
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Annexure: Code Snippets and Additional Resources

Thruster Controller Code

```
import rclpy
from rclpy.node import Node
from geometry_msgs.msg import Twist
import pigpio
import time
import sys

ESC_LEFT_PIN = 17
ESC_RIGHT_PIN = 18

PWM_NEUTRAL = 1500
PWM_MIN = 1100
PWM_MAX = 1900

def clamp(val, vmin, vmax):
    return max(min(val, vmax), vmin)

class ThrusterController(Node):
    def __init__(self):
        super().__init__('thruster_controller')
        self.pi = pigpio.pi()
        if not self.pi.connected:
            self.get_logger().error("Could not connect to pigpio daemon! Is it running?")
            sys.exit(1)

        self.get_logger().info("Connected to pigpio. Arming ESCs...")
        self.pi.set_servo_pulsewidth(ESC_LEFT_PIN, PWM_NEUTRAL)
        self.pi.set_servo_pulsewidth(ESC_RIGHT_PIN, PWM_NEUTRAL)
        time.sleep(3)
        self.get_logger().info("ESCs armed. Ready to receive commands.")

        self.subscription = self.create_subscription(
            Twist,
            '/thruster_cmd',
            self.listener_callback,
            10)

    def listener_callback(self, msg):
```



```

        left_pwm = clamp(int(msg.linear.x), PWM_MIN, PWM_MAX)
        right_pwm = clamp(int(msg.angular.z), PWM_MIN, PWM_MAX)

        self.pi.set_servo_pulsewidth(ESC_LEFT_PIN, left_pwm)
        self.pi.set_servo_pulsewidth(ESC_RIGHT_PIN, right_pwm)

        self.get_logger().info(
            f"Set Left PWM: {left_pwm} us | Right PWM: {right_pwm} us"
        )

    def destroy_node(self):
        self.get_logger().info("Shutting down, setting thrusters to neutral and
stopping PWM.")
        self.pi.set_servo_pulsewidth(ESC_LEFT_PIN, PWM_NEUTRAL)
        self.pi.set_servo_pulsewidth(ESC_RIGHT_PIN, PWM_NEUTRAL)
        time.sleep(1)
        self.pi.set_servo_pulsewidth(ESC_LEFT_PIN, 0)
        self.pi.set_servo_pulsewidth(ESC_RIGHT_PIN, 0)
        self.pi.stop()
        super().destroy_node()

def main(args=None):
    rclpy.init(args=args)
    node = ThrusterController()
    try:
        rclpy.spin(node)
    except KeyboardInterrupt:
        pass
    node.destroy_node()
    rclpy.shutdown()

if __name__ == '__main__':
    main()

```

Tele-operation Mode Code

```

import rclpy
from rclpy.node import Node
from geometry_msgs.msg import Twist
import sys, termios, tty, select, time, os

PWM_NEUTRAL = 1500
PWM_MIN = 1100
PWM_MAX = 1900
PWM_STEP = 40

```

```

class CustomTeleop(Node):
    def __init__(self):
        super().__init__('teleop_node')
        self.publisher = self.create_publisher(Twist, '/thruster_cmd', 10)
        self.left_pwm = PWM_NEUTRAL
        self.right_pwm = PWM_NEUTRAL
        self.last_straight_pwm = PWM_NEUTRAL # Last straight speed (forward or
reverse)
        self.settings = termios.tcgetattr(sys.stdin)

    def __del__(self):
        termios.tcsetattr(sys.stdin, termios.TCSADRAIN, self.settings)

    def get_key(self):
        tty.setraw(sys.stdin.fileno())
        rlist, _, _ = select.select([sys.stdin], [], [], 0.1)
        key = sys.stdin.read(1) if rlist else ''
        termios.tcsetattr(sys.stdin, termios.TCSADRAIN, self.settings)
        return key

    def display_status(self):
        os.system('clear')
        print("=== Thruster PWM Teleop (Incremental) ===")
        print("W: Increase both (forward)")
        print("S: Decrease both (backward)")
        print("A: Turn left (incremental)")
        print("D: Turn right (incremental)")
        print("R: Reset to neutral")
        print("Q: Quit")
        print(f"\nLeft PWM : {self.left_pwm} us")
        print(f"Right PWM: {self.right_pwm} us")
        print("=====")

    def publish_pwm(self):
        msg = Twist()
        msg.linear.x = float(self.left_pwm)
        msg.angular.z = float(self.right_pwm)
        self.publisher.publish(msg)

    def run(self):
        self.display_status()
        self.publish_pwm()
        try:
            while rclpy.ok():
                key = self.get_key()
                updated = False

```

```

if key == 'w':
    # Restore to last straight if turning, then increment
    if self.left_pwm != self.right_pwm:
        self.left_pwm = self.right_pwm = self.last_straight_pwm
    self.left_pwm = min(PWM_MAX, self.left_pwm + PWM_STEP)
    self.right_pwm = min(PWM_MAX, self.right_pwm + PWM_STEP)
    self.last_straight_pwm = self.left_pwm
    updated = True

elif key == 's':
    # Restore to last straight if turning, then decrement
    if self.left_pwm != self.right_pwm:
        self.left_pwm = self.right_pwm = self.last_straight_pwm
    self.left_pwm = max(PWM_MIN, self.left_pwm - PWM_STEP)
    self.right_pwm = max(PWM_MIN, self.right_pwm - PWM_STEP)
    self.last_straight_pwm = self.left_pwm
    updated = True

elif key == 'a':
    # Incremental left turn: left down, right up
    self.left_pwm = max(PWM_MIN, self.left_pwm - PWM_STEP)
    self.right_pwm = min(PWM_MAX, self.right_pwm + PWM_STEP)
    updated = True

elif key == 'd':
    # Incremental right turn: left up, right down
    self.left_pwm = min(PWM_MAX, self.left_pwm + PWM_STEP)
    self.right_pwm = max(PWM_MIN, self.right_pwm - PWM_STEP)
    updated = True

elif key in ('r', 'R'):
    self.left_pwm = PWM_NEUTRAL
    self.right_pwm = PWM_NEUTRAL
    self.last_straight_pwm = PWM_NEUTRAL
    updated = True

elif key in ('q', 'Q'):
    break

if updated:
    self.display_status()
    self.publish_pwm()
    time.sleep(0.05)

except KeyboardInterrupt:

```

```

        pass
    finally:
        self.left_pwm = PWM_NEUTRAL
        self.right_pwm = PWM_NEUTRAL
        self.publish_pwm()

def main(args=None):
    rclpy.init(args=args)
    node = CustomTeleop()
    node.run()
    node.destroy_node()
    rclpy.shutdown()

if __name__ == '__main__':
    main()

```

Thruster Controller Code

```

import rclpy
from rclpy.node import Node
from geometry_msgs.msg import Twist
import pigpio
import time
import sys

ESC_LEFT_PIN = 17
ESC_RIGHT_PIN = 18

PWM_NEUTRAL = 1500
PWM_MIN = 1100
PWM_MAX = 1900

def clamp(val, vmin, vmax):
    return max(min(val, vmax), vmin)

class ThrusterController(Node):
    def __init__(self):
        super().__init__('thruster_controller')
        self.pi = pigpio.pi()
        if not self.pi.connected:
            self.get_logger().error("Could not connect to pigpio daemon! Is it running?")
            sys.exit(1)

        self.get_logger().info("Connected to pigpio. Arming ESCs...")
        self.pi.set_servo_pulsewidth(ESC_LEFT_PIN, PWM_NEUTRAL)

```

```

self.pi.set_servo_pulsewidth(ESC_RIGHT_PIN, PWM_NEUTRAL)
time.sleep(3)
self.get_logger().info("ESCs armed. Ready to receive commands.")

self.subscription = self.create_subscription(
    Twist,
    '/thruster_cmd',
    self.listener_callback,
    10)

def listener_callback(self, msg):
    left_pwm = clamp(int(msg.linear.x), PWM_MIN, PWM_MAX)
    right_pwm = clamp(int(msg.angular.z), PWM_MIN, PWM_MAX)

    self.pi.set_servo_pulsewidth(ESC_LEFT_PIN, left_pwm)
    self.pi.set_servo_pulsewidth(ESC_RIGHT_PIN, right_pwm)

    self.get_logger().info(
        f"Set Left PWM: {left_pwm} us | Right PWM: {right_pwm} us"
    )

def destroy_node(self):
    self.get_logger().info("Shutting down, setting thrusters to neutral and
stopping PWM.")
    self.pi.set_servo_pulsewidth(ESC_LEFT_PIN, PWM_NEUTRAL)
    self.pi.set_servo_pulsewidth(ESC_RIGHT_PIN, PWM_NEUTRAL)
    time.sleep(1)
    self.pi.set_servo_pulsewidth(ESC_LEFT_PIN, 0)
    self.pi.set_servo_pulsewidth(ESC_RIGHT_PIN, 0)
    self.pi.stop()
    super().destroy_node()

def main(args=None):
    rclpy.init(args=args)
    node = ThrusterController()
    try:
        rclpy.spin(node)
    except KeyboardInterrupt:
        pass
    node.destroy_node()
    rclpy.shutdown()

if __name__ == '__main__':
    main()

```

Thruster Subscriber Code

```
import rclpy
from rclpy.node import Node
from geometry_msgs.msg import Twist

class ThrusterSubscriber(Node):
    def __init__(self):
        super().__init__('thruster_subscriber')
        self.subscription = self.create_subscription(
            Twist,
            '/thruster_cmd',
            self.listener_callback,
            10)
        self.get_logger().info('Thruster subscriber started, listening to /thruster_cmd')

    def listener_callback(self, msg):
        linear = msg.linear.x
        angular = msg.angular.z
        self.get_logger().info(f'Received: Linear={linear:.2f}, Angular={angular:.2f}')
        # TODO: Add your PWM output code here

def main(args=None):
    rclpy.init(args=args)
    node = ThrusterSubscriber()
    try:
        rclpy.spin(node)
    except KeyboardInterrupt:
        pass
    node.destroy_node()
    rclpy.shutdown()

if __name__ == '__main__':
    main()
```